

HARIPRASAD BATHINI SANKARAN

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DATA ENGINEER

SUMMARY

Data Engineer with **5+ years** architecting enterprise data platforms, ELT/ETL pipelines, and analytical frameworks across **Databricks**, **Snowflake**, **dbt**, and **Apache Airflow**. Expert in PySpark distributed processing, medallion (staging/marts) architectures, and **Kimball dimensional design** on **AWS** and **Azure**. Track record of processing **multi-terabyte daily telemetry**, unifying **50+ sources**, cutting latency **30%+**, and deploying reusable frameworks that shrink delivery timelines from weeks to days. Deep **Structured Streaming** and **Kafka** expertise, with domain depth in **cybersecurity** (SIEM, Splunk, Cribl), **healthcare (FHIR/HIPAA)**, and **GenAI/LLM** systems

SKILLS

Warehousing & ELT: Snowflake, dbt, SQL, ELT/ETL, Data Modeling (Kimball, Star & Snowflake Schema, Fact/Dimension), Data Warehousing, layered staging → marts architecture (medallion), Data Quality & Validation, Data Governance, Metadata-Driven Ingestion
Orchestration & Ingestion: Apache Airflow (DAG design & generators), Databricks Workflows, Apache Kafka, Structured Streaming, Cribl, REST API Ingestion (retry logic, pagination), AWS SQS event routing, Autoloader
Data Engineering: Python, PySpark, Apache Spark, Spark SQL, Data Structures & Algorithms, batch & real-time streaming pipelines
Cloud & DevOps: AWS (S3, Lambda, Glue, SQS, Redshift, Step Functions, Secrets Manager, KMS, IAM), Azure (Data Factory, ADLS Gen2, Azure SQL/SQL Server, Synapse, Key Vault, Entra ID), Terraform (IaC), CI/CD (Azure DevOps, GitHub Actions), Git/GitHub, Docker
Lakehouse: Databricks (Certified Data Engineer Professional), Delta Lake, Delta Live Tables, Unity Catalog, MLflow
Security & Compliance: SIEM (Splunk), CrowdStrike EDR telemetry, Cribl log routing, SOC enablement, RBAC / least-privilege, column-level security, dynamic data masking, PHI/HIPAA handling
Databases & Stores: PostgreSQL, DuckDB, ChromaDB, JSON, CSV, Parquet, Iceberg
GenAI & LLMs: RAG, LangChain, Ollama, vector embeddings, prompt engineering, AI-assisted development (Claude Code, Copilot, Gemini)
Business Intelligence: Power BI, Tableau, Dashboards, Operational Analytics, Reporting
Programming & Practices: Python, SQL, C++, Java, Unix/Bash, SDLC coding standards, Multiprocessing/Threading, Agile Methodology

CERTIFICATION

- **Databricks Certified Data Engineer Professional (Link)** Dec 2025
- **Azure Data Fundamentals (DP-900) (Link)** Mar 2022
- **Azure Fundamentals (AZ-900) (Link)** Jul 2021

WORK EXPERIENCE

Student Associate - Data Analyst | University of Cincinnati Jan 2026 – Present

- Provided supervisor with one reliable source of truth by building and maintaining **ETL pipelines** to a single reporting database.
- Delivered **Power BI** dashboards that turn raw package center metrics into at-a-glance operational views, day-to-day decisions for the team.
- Reduced report generation time by optimizing **SQL** queries and data models, giving stakeholders faster access to the metrics they act on.
- Reduced manual data-verification effort with automated **Python** validation scripts that catch integrity and consistency issues.

Data Engineer 2 - Cyber Security | Comcast Corporation Aug 2023 – Aug 2025

- Engineered enterprise security pipelines in **Snowflake** and **Databricks**, processing **multi-terabyte daily telemetry** into a centralized warehouse powering global network anomaly detection and event correlation.
- Reduced data processing latency **30%+** by orchestrating automated **ELT** workflows with **Apache Airflow** and Databricks Workflows on Terraform-provisioned, auto-scaling compute.
- Unified logs from **50+ security tools** into a single analytics layer, automating ingestion with **Cribl**, **Airflow**, and **AWS Lambda** (JSON → S3) and ending fragmented, tool-by-tool data access for analysts.
- Cut new data-source onboarding effort **40%** by building reusable ingestion templates and **Airflow DAG generators** the whole team adopted, delivering new analytics feeds in days instead of weeks.
- Sped threat investigations for downstream analysts by engineering a layered staging → intermediate → marts (**medallion**) architecture of **SQL-first transformations** delivering clean, tested, analytics-ready security data.
- Designed a log-routing architecture with **Cribl** ingesting high-volume telemetry from **Splunk**, plus custom **REST API** integrations pulling endpoint telemetry directly from **CrowdStrike**.
- Enabled threat-hunting teams to spot suspicious activity within minutes through **Structured Streaming** and **Kafka** pipelines for logs.
- Ensured data integrity across **10+ ELT pipelines** running up to **5 concurrent executions** with zero dependency failures through automated orchestration and dependency management.
- Cut warehouse and storage costs **\$5,000/year** through automated table-maintenance jobs, partition pruning, and clustering/compaction strategies for query-performance tuning.
- Achieved consistent, reproducible releases with zero manual intervention by implementing **Terraform** modules for **CI/CD** for deploying jobs, warehouses/clusters, and permissions across dev, staging, and production.
- Accelerated ELT development cycles **30%** by integrating AI-assisted tools (**GitHub Copilot**, **Gemini**, **Claude Code**) into development, validation, and documentation.

Data Engineer | Infosys Ltd. (Client: Microsoft Corporation)

Jun 2021 – Aug 2023

- Reduced new data-source integration time **50%** organization-wide by building a reusable **Python REST API** ingestion notebook (retry logic, pagination) that other teams adopted for external data onboarding.
- Enabled self-service analytics and executive dashboards by designing dimensional **star and snowflake schema** data models (facts/dimensions) in the cloud data warehouse for reliable, decision-ready reporting.
- Cut daily batch runtime by **30+ minutes** and improved performance **15%** by converting legacy **SQL/T-SQL stored procedures** into modular, version-controlled transformation frameworks in PySpark.
- Increased pipeline scalability and maintainability with a **metadata-driven ingestion framework** in ADF automating source-to-target mapping, schema validation, and dynamic orchestration.
- Improved data consistency and refresh accuracy **40%+** by engineering large-scale ELT in **Azure Databricks** and **ADF** that ingests and aggregates multi-source data into **ADLS Gen2**.
- Accelerated transformations across **billions of records** by tuning partitioning, caching, and shuffle strategies for high-performance jobs.
- Reduced maintenance windows **30 min/run** via automated table lifecycle management using Python **multiprocessing** to parallelize vacuum and optimize operations.
- Hardened business-critical feeds with rigorous **data quality and validation checks** (including automated schema-drift detection in Python) across all ingestion layers.

Software Engineer - Data | Zoho Corporation Pvt. Ltd.

Jan 2020 – Dec 2020

- Improved report accuracy and cut data-request turnaround by authoring and optimizing complex **SQL** for BI reports used by product teams.
- Reduced manual effort in recurring workflows with **Python** scripts for data cleaning, preprocessing, and file automation.
- Supported automation of key business metrics and dashboards by building and maintaining SQL-driven pipelines for internal reporting.
- Collaborated with cross-functional teams to translate data requirements into structured extraction and transformation processes.

PROJECTS

Hybrid Healthcare Data Engineering Pipeline (FHIR → Warehouse Star Schema → RBAC) | Personal Project

[Link](#) 

- Engineered a layered staging → intermediate → marts (Bronze → Silver → Gold) pipeline ingesting synthetic **FHIR R4** bundles with **PySpark**, using dynamic resource fan-out to auto-discover and build dedicated staging tables for all **19 FHIR resource types**, including Claims, Patients, Encounters, and Conditions.
- Modeled a Kimball-compliant **star schema (2 fact, 9 dimension tables)** in the warehouse serving layer using deterministic **MD5 surrogate keys** to maintain referential integrity across complex FHIR reference patterns spanning Claim, Condition, and Procedure resources.
- Combined batch and real-time streaming (**Python Kafka Producer** → **Kafka on Docker** → **Structured Streaming**) to simulate concurrent hospital feeds across **50 partitioned data folders**.

Security Log Analytics and Threat Detection Platform (Azure) | Personal Project

[Link](#) 

- Built an end-to-end security-telemetry analytics platform on **ADF** and **Databricks**, processing firewall, endpoint, and network logs from ADLS Gen2 into SIEM-ready curated datasets via a layered staging → marts (medallion) architecture.
- Automated infrastructure and scheduling with **Terraform**, **Azure DevOps**, and **Apache Airflow** for version-controlled, SLA-compliant, fully auditable production runs.
- Developed **PySpark** and **SQL** transformations to clean, enrich, and structure raw JSON security events into Parquet/Delta datasets, with schema validation and data-quality checks that flag failed loads pre-consumption.
- Delivered **Power BI** and Synapse dashboards surfacing real-time threat anomalies, top attack sources, and detection metrics for cybersecurity teams.

Local RAG Pipeline: Healthcare Patient Data Q&A with LangChain and Ollama | Personal Project

- Built an end-to-end **RAG** pipeline using **LangChain** and locally hosted **Gemma 4 (12B)** via **Ollama**, enabling fully offline, privacy-preserving Q&A over structured healthcare data without external API dependencies.
- Engineered a structured-to-text transformation layer converting **Parquet** records (output of an ETL pipeline) into semantically rich text summaries, making structured patient data searchable via natural language.
- Built a vector ingestion pipeline using **nomic-embed-text** embeddings and **ChromaDB** as the local vector store, supporting batch ingestion of multi-part Parquet datasets with persistent storage.
- Implemented an interactive conversational interface that retrieves semantically relevant patient records from ChromaDB and passes them as grounded context to Gemma 4, reducing hallucinations and anchoring responses to real data.

EDUCATION

University of Cincinnati

Master of Science in Information Technology